

Resume Pilot

Izabela Camaj

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Senior Marketing Manager 7+ years of social marketing experience, driving customer growth and engagement in digital, B2B, and content marketing campaigns. Increased brand awareness by 20%, mobile traffic by 40%, customer acquisition by 25%, customer lifetime value by 15%, and sales revenue by 30% in marketing and campaign strategies, resulting in a 45% increase in lead conversion.	SKILLS Advertising, Analytics, CRM, Marketing, B2B, Business Management, B2C, Business Management, Content Marketing, Cross Channel Campaigns, Customer Acquisition, Customer Lifetime Value, Data Analysis, Digital Advertising, Digital Marketing, Email Marketing, Events Management, Google Analytics, Lead Conversion, Lead Generation, Management, Marketing Campaign Management, Marketing Technology, Market Tools, Media Campaigns, Mobile Strategy, New Business Development, Paid Digital Campaigns, Partner Management, PPC, Public Safety, Product Marketing, Project Management, Social Media Management, Sales Management, Sales Strategy, Social Media, Strategy, Social Marketing, Web Traffic, YouTube
Work History MetLife • New York, United States • 07/2024 - Present Senior Marketing Manager - Increased lead generation by 45% in 6 months through the development and execution of cross-channel campaigns targeting key customer segments. - Increased the company's online presence by 25%, driving a 40% increase in website traffic and generating \$2M in revenue through successful digital campaigns. - Launched and managed over 15 marketing campaigns, uncovering actionable insights that led to a 25% increase in customer acquisition and a 40% growth in customer lifetime value over one year.	
MetLife • 05/2023 - 06/2023 Marketing Manager - Managed a comprehensive cross-functional marketing campaign for a new product launch, generating a total of 150,000 leads, coordinating promotional activities, and executing digital marketing strategies across multiple channels. The campaign resulted in a 20% increase in brand awareness, generating \$500,000 in sales revenue within 30 days. - Implemented a client loyalty program strategy, resulting in a 15% increase in repeat sales and a 10% increase in customer lifetime value.	
White Lotus Resort • 01/2021 - 01/2020 Marketing Manager - Increased digital advertising revenue by 35% within 6 months by leveraging strategic B2B and B2C relationships, enhancing client engagement. - Developed a comprehensive B2B/B2C digital advertising strategy, driving a 300% revenue increase within 6 months by optimizing targeted campaigns.	

Miranda Loper 1234 Wagon Street Miami, FL 33101 (305) 111-1234 mloper@cs5130.com	
Summary:	More than 10 years of experience as a real estate paralegal and contracts manager for nationally recognized law firms and wireless communications companies.
Contract Management:	Prepared and managed legal documents required to protect client digital, residential, utility and wireless assets, and protection of intellectual property. Conducted research and due diligence, resulting in a 15% increase in revenue growth. Coordinated in-house and outside counsel regarding litigation for more than 25% of active suits.
Real Estate Paralegal:	Coordinated review and obtained legal approval for all regional leases and lease amendments related to site alterations. Served as liaison during the cultivation process. Prepared and supervised maintenance databases for lease tracking and property management.
Property Management:	Served as director of all facilities, maintaining legal documents and mitigating site issues for 18 international locations.
Leadership:	Directly managed 180 Legal Department, comprised of 50 onsite staff. Served as Mid-South Area Real Estate Coordinator, overseeing 25 employees at five sites.
Recruiting & Landing:	Drafted loan documents, review instructions, and title obligation letters. Conducted due diligence on title and survey matters. Organized and attended closings.
Work History	2016 - Present WILLIAMS COMMUNICATIONS INC., Raleigh, NC Contract Manager, Facilities Database 2010 - 2016 WILLIAMS COMMUNICATIONS, Cary, NC Real Estate Specialist, Mid-South Network 2009 - 2010 BUILDING & FINANCE PARTNERS DANIEL HENDERSON, JAMISON & WALKER LLP, Murphy, NC Real Estate Paralegal 2004 B.A., POLITICAL SCIENCE University of North Carolina, Chapel Hill, NC 2000 CERTIFICATES OF ACADEMIC ACHIEVEMENT Duke University (Cum Laude), Durham, NC Additional Skills: Microsoft Office Suite 2010, Lotus Notes 2000 Retirement System, Broad Autism System v4.0, Soundstage, iCaption, iCaption Legal Billing System.

CSI 5130

Introduction

- Motivation
 - Internships applications
- Users
 - Students seeking internships
 - Adults seeking professional jobs
 - Employees seeking jobs at other companies
- Resume Analysis
- Suggestions & Notes
 - Career prediction
 - Resume improvements
 - Skill recommendation
 - Resume score
 - Highlights
 - Keywords



Design

Resume Pilot

Scan Attatchements

Choose File

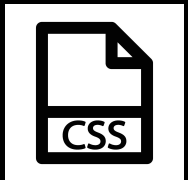
Analyze

Scan Results

Tools & Packages

- Visual Studio Code
- Python
 - FastAPI
 - AI Models
- CSS
- HTML
- Javascript

- .venv
- fastapi uvicorn
- pdfminer.six
- pandas
- scikit-learn
- torch
- pillow
- python-multipart



Method: Probabilistic Classification

- **Trained on thousands of labeled resumes**
 - The model learns patterns between resume content and job categories (e.g., “software”, “medical”, “finance”)
- **Outputs probability for each career field**
 - Instead of a single prediction, it gives a probability distribution over all fields
 - Lets you report both **predicted field** and **confidence score**

Method: Markov Decision Process

- **Represents a resume as a sequence of states**
 - States = key resume sections
 - Captures the structure of typical resume progression
- **Actions represent improvements**
 - “Add”, “Improve”, or “Skip” for each section
- **MDP transitions model how a user should improve their resume**
 - After evaluating one section, the model moves to the next predictable state
- **Used to generate actionable suggestions**
 - Example output: “Objective: add”, “Skills: improve”, etc
 - Helps users prioritize improvements

Method: Deep Q-Learning

- **Idea: model skill selection as a reinforcement learning problem**
 - State = resume text
 - Actions = possible skills to add
 - Rewards = how much adding a skill improves the resume (conceptually inspired)
- **Your implementation uses a learned skill map instead of a neural network**
 - Extracts top skills per career field from dataset
 - Compares them to the resume to find missing skills
- **Approximates Q-values using frequency and relevance**
- **Used to recommend personalized missing skills**
 - E.g., “Python”, “Excel”, “Research”, “Communication”, etc

Method: Policy Gradient

- **Evaluates two things:**
 - **Section coverage** - how complete the resume structure is
 - Objective, skills, projects, achievements, education
 - **Length appropriateness** - compares word count to dataset average
- **Combines features into a final score (0–100)**
 - Weighted sum - Reward aggregation from policy gradient methods
- **Used in to give a quality score**
 - Helps users understand if their resume is too short, too long, or missing sections

Method: Self Attention (PyTorch)

- **Computes attention weights between words in the resume**
- **Transforms text into embeddings**
 - Each token ID maps to a dense vector representation
 - Allows numerical comparison of words
- **Attention mechanism highlights important tokens**
 - Heads focus on keywords like “Python”, “Research”, “Clinical”, etc
- **Used to produce “Keywords”**
 - Helps users see what stands out in their resume

Experimental Results

- **index.html**
 - Hosts UI layout & upload form
 - Loads CSS and JS
- **styles.css**
 - Handles all styling for the interface, layout, buttons, and scrollable result sections
- **script.js**
 - Sends the resume file to the FastAPI backend (/analyze)
 - Displays loading states, final results, and error messages
 - Updates UI dynamically with the returned JSON

Experimental Results

- Single entry-point file: app.py
- Responsibilities:
 - Serves the frontend files (index.html, CSS, JS)
 - Provides the /analyze API endpoint
 - Saves uploaded resumes to data/resumes_uploaded/
 - Extracts text from PDFs using pdfminer.six
 - Passes extracted text into each AI models
 - Returns a JSON object with:
 - Career prediction
 - Resume improvements
 - Skill recommendation
 - Resume score
 - Highlights
 - Keywords

Challenges

- Setting goals
- Working in Visual Studio Code
- Connecting the Resume dataset
 - Discovery when clicking the “Analyze” button for the same file
- Adding an additional feature
 - Scanning job post requirements and comparing it against that
- Uploaded resume being saved in the folder
- Installing necessary packages & Working with FastAPI
- Difficulty with Streamlit and Pyresparsers
- Skills & Keywords return

DEMO

Thank You

Questions?